For Examiner's Use

6 An electron travelling horizontally with a velocity of 9.0×10^6 m s^{-1} enters perpendicularly into a uniform electric field between two parallel metal plates as shown in Figure 6.1. After leaving the electric field, it travels into a region of uniform magnetic field of magnetic flux density 0.72 mT. The magnetic field is directed normally into the plane of the paper.

Assume that viscous drag and gravitational force are negligible.

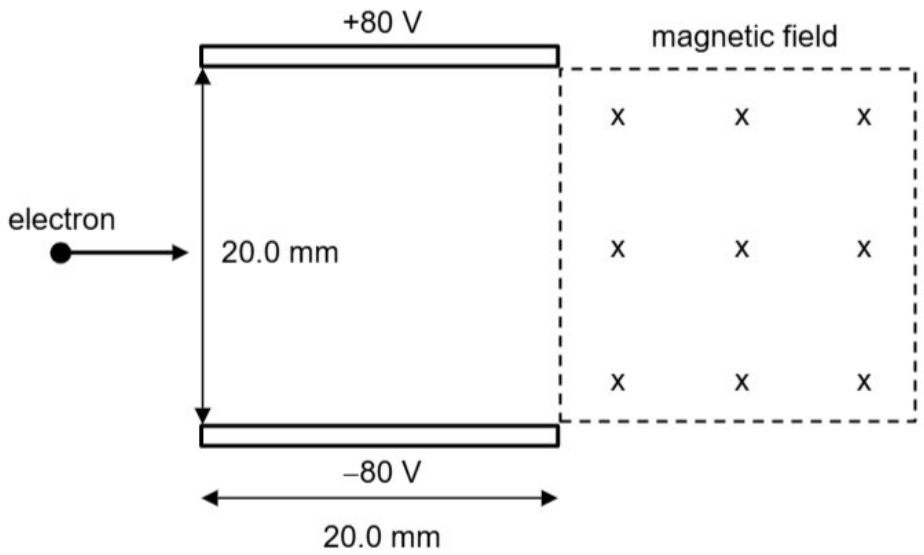


Figure 6.1

(a) Determine the time taken for the electron to travel through the electric field.

time = ... s [1]

(b) Show that the velocity of the electron just before it enters the magnetic field is 9.5×10^6 m s^{-1}.

[3]

(c) Determine the radius of the path taken by the electron when it enters the magnetic field.

radius = ... m [2]

(d) On Figure 6.1, sketch a possible path taken by the electron in the electric and magnetic fields. [2]

[Total: 8]